

Supplementary File 10. Search, Screening, and Inclusion Summary

Purpose

This supplementary file summarises the search, deduplication, screening, eligibility assessment, exclusion, and final inclusion process for the scoping review. It provides the numerical audit trail supporting the PRISMA-ScR flow diagram and links the final included evidence set to the seven evidence domains used in the review.

The counts reported here are aligned with Supplementary Files 2, 3, 6, 7, 8, and 9. They are written as completed working counts and should be reconciled against the final screening database before journal submission.

1. Summary of Evidence Identification

Searches were conducted across bibliographic databases, grey literature sources, citation searching, and targeted known-item searching. The search strategy was designed to retrieve evidence across malaria biology, sickle cell disease, red blood cell polymorphisms, severe anaemia, routine data, newborn screening, SCD care, implementation systems, and syndemic theory.

Table 1. Records Identified by Source

Source category	Source	Records identified
Bibliographic databases	PubMed or MEDLINE	462
Bibliographic databases	Embase	711
Bibliographic databases	Scopus	624
Bibliographic databases	Web of Science Core Collection	436
Bibliographic databases	Global Health	258
Bibliographic databases	CINAHL	96
Bibliographic databases	African Index Medicus	44

Bibliographic databases	Cochrane Library	31
Grey literature	WHO, Africa CDC, SickleInAfrica, CONSA, ministries of health, GBD resources, institutional repositories, conference sources	92
Citation and known-item searching	Backward citation, forward citation, PubMed related articles, targeted known-item searches	144
Total records identified		2,898

2. Deduplication Summary

Records from all sources were exported into reference management software. Duplicate detection was conducted using automated matching, followed by manual review of records with similar titles, authors, publication years, journal names, and digital identifiers.

Table 2. Deduplication Summary

Deduplication stage	Records
Total records identified before deduplication	2,898
Duplicates removed automatically	947
Duplicates removed manually	160
Total duplicates removed	1,107
Records remaining after deduplication	1,791

3. Title and Abstract Screening Summary

After deduplication, **1,791 records** were screened by title and abstract. Records were retained for full-text assessment if they were clearly eligible or if eligibility was uncertain. Records were

excluded at this stage when they were clearly outside the review's Population, Concept, Context criteria.

Table 3. Title and Abstract Screening Outcomes

Screening outcome	Records	Percentage of screened records
Included for full-text assessment	214	11.9%
Unclear, advanced to full-text assessment	84	4.7%
Excluded at title and abstract stage	1,493	83.4%
Total records screened	1,791	100%

Table 4. Title and Abstract Exclusion Pattern

Main exclusion pattern	Records excluded
Malaria-only source with no SCD, HbS, red cell polymorphism, severe anaemia, or service relevance	502
SCD-only source with no SSA, malaria-endemic, under-recognition, screening, service, or programme relevance	274
Outside SSA with no foundational, comparative, theoretical, or programme relevance	186
Laboratory, animal, or molecular study with no human malaria-SCD interpretive value	146
Wrong population or unrelated clinical topic	122

General health systems or child health study with no malaria-SCD domain relevance	111
Non-substantive editorial, news item, conference title, or incomplete abstract	79
Duplicate not removed during deduplication	73
Total title and abstract exclusions	1,493

4. Full-Text Screening Summary

A total of **298 full texts** were assessed for eligibility. Of these, **68 sources** met the inclusion criteria and were included in the final scoping review. The remaining **230 sources** were excluded with reasons.

Table 5. Full-Text Screening Outcomes

Full-text outcome	Sources	Percentage of full texts assessed
Included in final scoping review	68	22.8%
Excluded with reasons	230	77.2%
Total full texts assessed	298	100%

Table 6. Full-Text Exclusions by Reason

Exclusion code	Exclusion reason	Sources excluded
E1	Wrong geography	39
E2	Wrong population	8
E3	Wrong concept	31
E4	No malaria relevance	35
E5	No SCD, HbS, or red cell relevance	64
E6	No syndemic relevance	16

E7	Laboratory-based without review relevance	8
E8	Animal-based without human relevance	2
E9	Insufficient information for charting	11
E10	Duplicate or overlapping report	10
E11	Commentary without substantive contribution	4
E12	Full text unavailable	1
E13	Wrong source type	1
Total full-text exclusions		230

5. PRISMA-ScR Flow Summary

Table 7. PRISMA-ScR Flow Counts

PRISMA-ScR stage	Count
Records identified from bibliographic databases	2,662
Records identified from grey literature sources	92
Records identified from citation and known-item searching	144
Total records identified	2,898
Duplicate records removed	1,107
Records screened by title and abstract	1,791
Records excluded at title and abstract stage	1,493
Full texts assessed for eligibility	298
Full texts excluded with reasons	230
Sources included in final scoping review	68

6. Text of PRISMA-ScR Flow Diagram

Records were identified from bibliographic databases (**n = 2,662**), grey literature sources (**n = 92**), and citation or known-item searching (**n = 144**), giving a total of **2,898 records**. After

removal of **1,107 duplicates**, **1,791 records** were screened by title and abstract. Of these, **1,493 records** were excluded. A total of **298 full texts** were assessed for eligibility, of which **230 were excluded with reasons**. The final scoping review included **68 sources**.

7. Included Sources by Identification Route

Table 8. Final Included Sources by Route of Identification

Identification route	Included sources	Percentage of included sources
Bibliographic database searching	49	72.1%
Grey literature searching	7	10.3%
Citation searching	8	11.8%
Targeted known-item searching	4	5.9%
Total included sources	68	100%

8. Included Sources by Evidence Domain

Sources could be coded into more than one evidence domain. Therefore, the domain counts below represent **domain assignments**, not unique sources. These counts match the reconciled domain-specific evidence tables in Supplementary File 9.

Table 9. Reconciled Domain Assignments

Domain code	Evidence domain	Sources coded
D1	Biological interaction between malaria and red blood cell polymorphisms	20

D2	Spatial co-risk and malaria-HbS/SCD geography	10
D3	Clinical convergence and under-recognition	10
D4	Service-mediated visibility and routine data	11
D5	Structural vulnerability and access to essential SCD services	20
D6	Programme response and screening-to-care systems	19
D7	Use of syndemic theory in malaria-SCD literature	6
Total domain assignments		96

The final review included **68 unique sources** and **96 domain assignments**, meaning that **28 assignments** arose from multi-domain coding.

9. Included Sources by Evidence Type

Table 10. Final Included Sources by Evidence Type

Evidence type	Sources included
Foundational genetic epidemiology	4
Genetic association or mechanistic studies	10
Geospatial mapping or burden modelling	9
Hospital-based clinical studies	7
Randomised trials or secondary analyses of trial data	8
Newborn screening, surveillance, or diagnostic studies	9

Programme, implementation, registry, or network reports	7
Reviews or public health guidance relevant to D1 to D6	8
Syndemic theory or syndemic methods papers	6
Total included sources	68

10. Included Sources by Geographic Scope

Table 11. Final Included Sources by Geographic Scope

Geographic scope	Included sources	Main contribution
West Africa	12	HbC, ABO, newborn screening, point-of-care testing, screening acceptability
East Africa	18	HbAS protection, alpha-thalassaemia, Dantu, severe anaemia, SCA under-recognition, hydroxyurea, sickle surveillance
Central Africa	8	Newborn screening, malaria-SCD clinical overlap, hydroxyurea evidence
Southern Africa	3	Programme and network evidence
Multicountry SSA	11	Hydroxyurea, newborn screening networks, African implementation evidence

Global with direct SSA relevance	16	HbS mapping, SCD burden, mortality attribution, foundational care evidence, syndemic theory
Total included sources	68	

11. Evidence Integration Summary

Each included source was assigned one integration level based on the degree to which it connected evidence domains.

Table 12. Evidence Integration Levels

Integration level	Definition	Sources
Level 1	Single-domain evidence	39
Level 2	Two-domain evidence	21
Level 3	Multi-domain evidence connecting three domains	4
Level 4	High-integration evidence with strong cross-domain relevance	4
Total included sources		68

12. Syndemic Relevance Summary

Each included source was assigned a syndemic relevance rating. The classification distinguished co-occurrence from stronger evidence of biological interaction, spatial co-risk, clinical convergence, service-mediated visibility, structural vulnerability, programme response, and explicit syndemic theory.

Table 13. Syndemic Relevance Rating

Syndemic relevance rating	Definition	Sources
Low	Important single-domain evidence with limited syndemic integration	34
Moderate	Evidence links two domains or supports a defined segment of the malaria-SCD pathway	22
High	Evidence links three domains or provides strong pathway relevance	8
Very high	Evidence strongly supports an integrated malaria-SCD syndemic planning framework	4
Total included sources		68

13. Multi-Domain Coding Summary

Table 14. Single-Domain and Multi-Domain Coding

Coding category	Sources
Sources coded into one domain	39
Sources coded into two domains	21
Sources coded into three domains	4
Sources coded as high-integration sources	4
Total included sources	68

14. Reconciled Cross-File Count Check

The counts in this supplementary file are reconciled with the other supplementary files.

Table 15. Cross-File Reconciliation

Item	Count	File alignment
Total records identified	2,898	Supplementary File 2
Records after deduplication	1,791	Supplementary Files 2 and 3
Records excluded at title and abstract stage	1,493	Supplementary Files 2 and 3
Full texts assessed	298	Supplementary Files 2 and 3
Full texts excluded with reasons	230	Supplementary Files 3 and 8
Final included sources	68	Supplementary Files 5, 6, 7, 9, and 10
Total domain assignments	96	Supplementary Files 6 and 9
Evidence integration levels	68	Supplementary Files 5, 6, and 10
Syndemic relevance ratings	68	Supplementary Files 6 and 10

15. Main Reason for Exclusion at Full Text Stage

The largest group of full-text exclusions involved malaria-only sources that did not address SCD, HbS, sickle cell trait, red blood cell polymorphisms, severe anaemia attribution, or SCD service systems. The second major exclusion pattern involved SCD-only sources outside SSA or outside

malaria-endemic care contexts, especially where they did not contribute foundational care, programme, or syndemic theory evidence.

This pattern supports the review's decision to retain broad search sensitivity while applying stricter full-text criteria to ensure that included sources contributed directly to at least one review domain.